α-labeling of banana tree graphs

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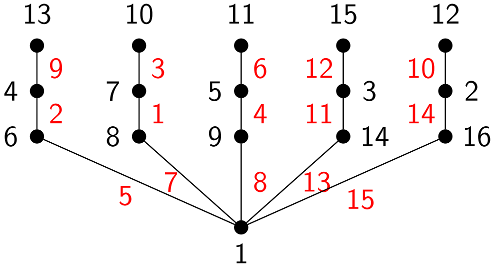
Sections

Motivation

- Focus: α-**labeling** and a special family of trees – banana trees
- Rosa (1967) introduced graceful (β) labeling and α-labeling
- Ringel-Kotzig conjecture: All trees are graceful
- Applications of α-labelings: coding theory, graph decompositions

Graceful labeling

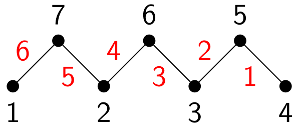
- $f : V \to \{1, 2, ..., m + 1\}$
- $f^*(uv) = |f(u) - f(v)|,\ uv \in E$
- f^* must be a bijection to $\{1, ..., n - 1\}$



Graceful labeling of a tree

α-labeling

- Additional property: there exists α such that for each edge uv exactly one inequality holds:
  1. $f(u) \leq \alpha < f(v)$
  2. $f(v) \leq \alpha < f(u)$



α-labeling of a tree

Conjugate and reverse α-labelings

Remark

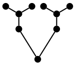
1. If f is an α-labeling of G and $f^*(vw) = 1$ for some $v, w \in V$, then $\alpha = \min\{f(v), f(w)\}$.
2. If f is an α-labeling of a graph G with n vertices, then its complementary labeling $\overline{f}$ defined as $\overline{f}(v) = n + 1 - f(v)$ is also an α-labeling of G with $\overline{\alpha} = n - \alpha + 1$.
3. If f is an α-labeling of G, its reverse (or inverse) labeling f^r defined as

$$f^r(v) = \begin{cases} \alpha + 1 - f(v), & f(v) \leq \alpha \\ n + \alpha + 1 - f(v), & f(v) > \alpha \end{cases}$$

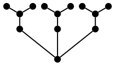
is also an α-labeling of G with $\alpha^r = \alpha$.

Definition of a banana tree $B(n, k)$

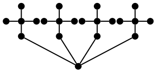
- Hierarchical structure: n star graphs S_k
- Each star is connected to a central vertex (root)



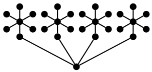
$B(2, 4)$



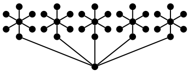
$B(3, 4)$



$B(4, 5)$



$B(4, 7)$



$B(5, 7)$

RSF and isomorphic labelings

- $\mathcal{L}_i = \{l_{i,j} \mid j \in [k-2]\}$ and $\mathcal{B}_i = \{a_i\} \cup \mathcal{L}_i$
- swapping the labels of two leaves or two stars doesn't change the labeling

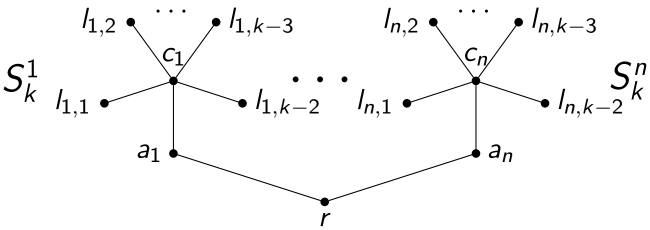


Figure: Rooted standard form of a $B(n,k)$ tree

The set of labelings $B(n, k)$

Definition

Let $\mathcal{A}(n, k)$ be the set of all α-labelings of the $B(n, k)$ tree and $\mathcal{A}_x(n, k)$ be the set of all α-labelings of the $B(n, k)$ tree where $f(r) = x$. Let the set of principal α-labelings $\mathcal{P}(n, k)$ be defined as $\mathcal{P}(n, k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n, k)$. Lastly, let the set of *central* α-labelings $\mathcal{C}(n, k)$ be defined as $\mathcal{C}(n, k) = \mathcal{A}_{\lfloor \frac{n}{2} \rfloor + 1}(n, k)$.

Proposition

Let f be an α-labeling of the $B(n, k)$ tree for any $n \in \mathbb{N}$ and $k \geq 2$. Then,

$$f(r) \leq \alpha \iff f(c_i) \leq \alpha \quad \forall i \in [n]$$

Partitioning $\mathcal{A}(n, k)$

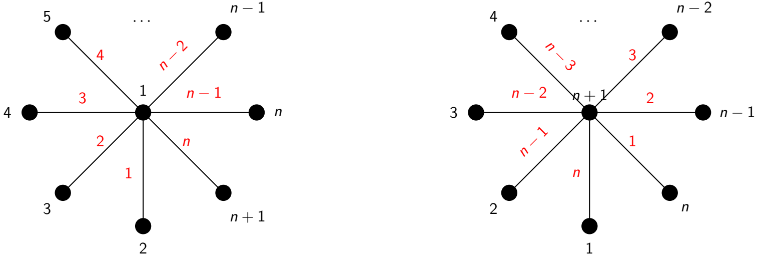
Theorem

For every $n, k \in \mathbb{N}$, we have

$$\mathcal{A}(n,k) = \mathcal{P}(n,k) \cup \mathcal{P}^r(n,k) \cup \overline{\mathcal{P}}(n,k) \cup \overline{\mathcal{P}^r}(n,k),$$

where $\mathcal{P}^r(n,k) = \{f^r : f \in \mathcal{P}(n,k)\}$, $\overline{\mathcal{P}}(n,k) = \{\overline{f} : f \in \mathcal{P}(n,k)\}$ and $\overline{\mathcal{P}^r}(n,k) = \{\overline{f^r} : f \in \mathcal{P}(n,k)\}$.

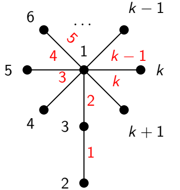
- Reducing the set of possible α-labelings by a factor of 4
- Reducing the complexity of the proofs



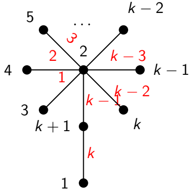
α-labeling f_1 of $B(n,1)$ α-labeling f_2 of $B(n,1)$

Figure: Two α-labelings of $B(n,1)$

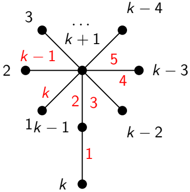
Classifying all the α-labelings of all the $B(1, k)$ trees



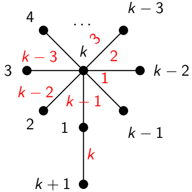
α-labeling f_1 with $\alpha_1 = 2$



α-labeling f_2 with $\alpha_2 = 2$



α-labeling f_3 with $\alpha_3 = k - 1$



α-labeling f_4 with $\alpha_4 = k - 1$

Classifying all the α-labelings of all the $B(2, k)$ trees

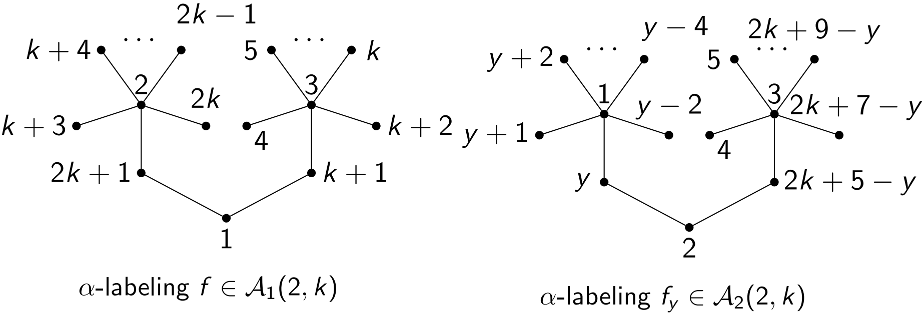
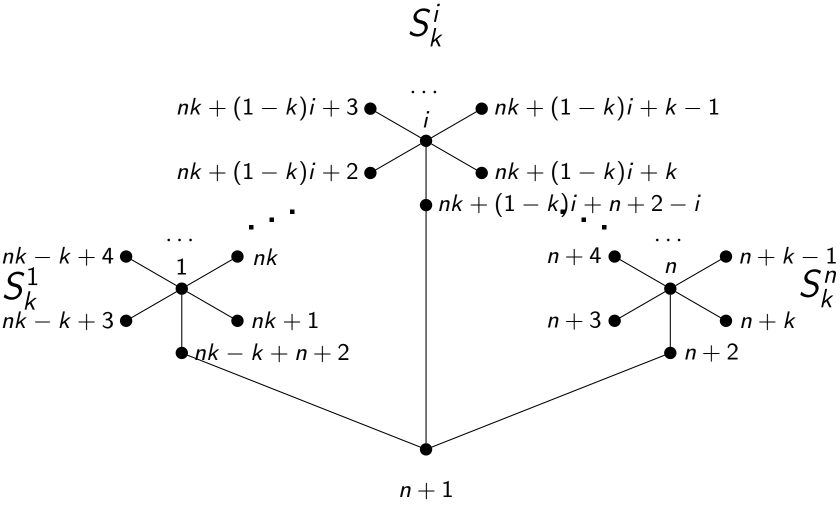


Figure: Two types of principal α-labelings of $B(2, k)$ (with $k + 3 \le y \le 2k + 1$)

A sketch of the proof for $B(n, k)$ for $k \geq n$



A sketch of the proof for $B(n, k)$ for $\frac{n}{2} + 1 \leq k \leq n$

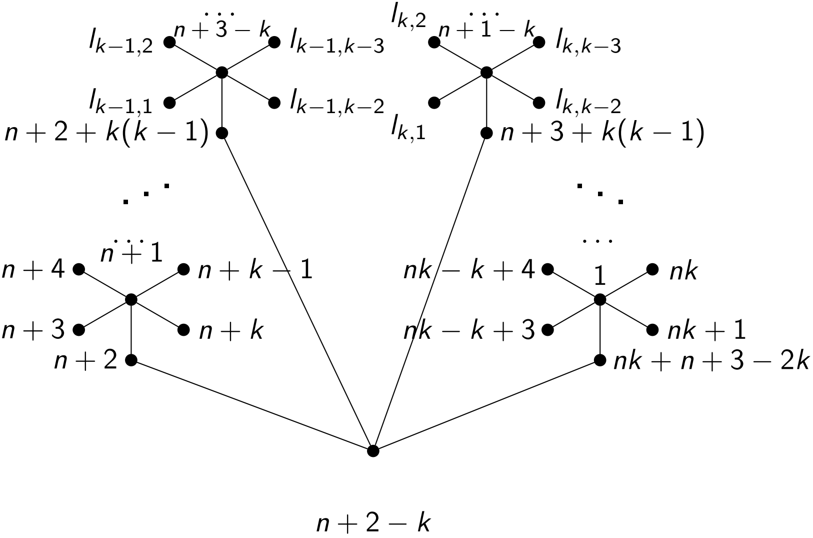
$$f(r) = n + 2 - k$$

$$f(c_i) = \begin{cases} n + 2 - i, & i < k \\ n + 1 - i, & i \geq k \end{cases}$$

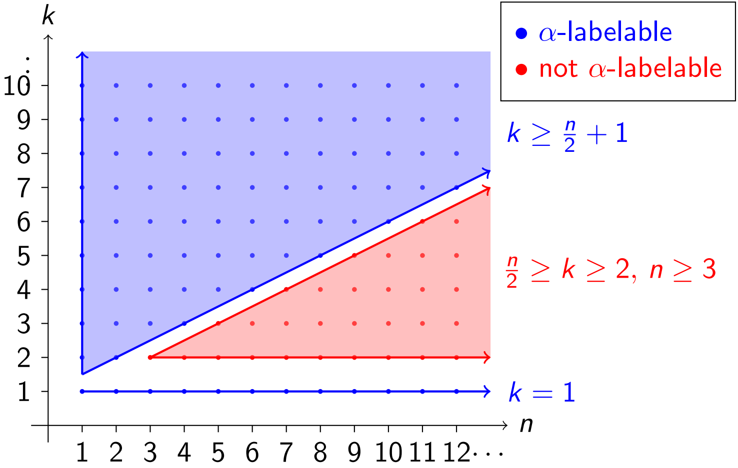
$$f(a_i) = \begin{cases} n + 2 - k + ki, & i < k \\ n + 3 - 2k + ki, & i \geq k \end{cases}$$

$$\mathcal{F}(\mathcal{L}_i) = \{n + 3 - k + (k - 1)i, \ldots, n + 1 + (k - 1)i\} \setminus \{f(a_i)\}$$

A sketch of the proof for $B(n,k)$ for $\frac{n}{2}+1 \le k \le n$



α-labelability of $B(n,k)$ trees



Conclusion

- We settled the open question of the existence of α-labelings for $B(n,k)$ trees for all $n,k \in \mathbb{N}$
- Introduced new notation and concepts for α-labelings of $B(n,k)$ trees
- Opened new questions regarding the number of non-isomorphic α-labelings of $B(n,k)$ trees for $n \geq 3$

Thank you for your attention!

Any questions?